Smart Lab Equipment & Slot Reservation Portal

Software Architecture

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Overview

The portal helps students and faculty check equipment availability, reserve slots, and return equipment. Lab administrators can manage equipment, calibration, and reports. The system is organized into three layers so that each part has a clear responsibility.

Why Layered Architecture?

I selected layered architecture because the system has different tasks that can be separated into the user interface, the services that handle rules, and the database. This makes the system easier to maintain and test. It also allows login and role checks to be handled before users access restricted functions. Keeping service logic separate from the interface helps organize requests and database operations.

Main Layers

Layer	What it does
Presentation Layer	Contains the User Interface, Admin Interface, and API Gateway. Users can log in, view equipment, reserve slots, return equipment, and view reports. The gateway routes requests to the required service and checks authentication.
Application / Business Logic Layer	Contains Authentication, Equipment, Reservation, Conflict Check, Return Management, and Calibration services. These services handle permissions, availability, double-booking checks, returns, late fees, and calibration reminders.
Data Layer	Contains the Smart Lab Database. It stores user details, equipment, slots, reservations, returns, and calibration records. Services read or update the data when needed.

A student opens the portal and sends a reservation request. The request goes from the interface to the API Gateway and then to the Reservation Service. The Conflict Check Service checks whether the slot is free. If it is available, the reservation is saved in the database and the result is shown to the student.

Database Information

The database has six main groups: Users, Equipment, Slots, Reservations, Returns, and Calibration Records. These records help the portal track who booked equipment, which slot was booked, return details, and calibration status.

Conclusion

Layered architecture is suitable for this portal because it separates the interface, system rules, and data storage. This structure keeps the design clear and makes the main functions easier to manage.